

Thursday Warm-up: The Auxiliary Field $\mathbf{H}$, and Linear Media

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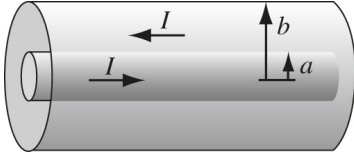


Figure 1: A coaxial cable system with magnetic susceptibility χ_m between the inner and outer conductors.

- (b) As a check, calculate the magnetization and the bound currents, and confirm that (together with the free currents) they generate the correct field.

1 Memory Bank

- **Bound volume current density:**

$$\mathbf{J}_b = \nabla \times \mathbf{M} \quad (1)$$

- **Bound surface current density:**

$$\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}} \quad (2)$$

- **Auxiliary field, $\mathbf{H}$:**

$$\mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M} \quad (3)$$

- **Magnetic linear media:**

$$\mathbf{M} = \chi_m \mathbf{H} \quad (4)$$

$$\mathbf{B} = \mu \mathbf{H} \quad (5)$$

$$\mu = \mu_0 (1 + \chi_m) \quad (6)$$

2. A free current I flows down a long straight wire of radius a . The wire is made of linear material with susceptibility χ_m , and the current is distributed uniformly.

- What is the $\mathbf{B}$ -field a distance s from the axis?

- Find all the bound currents.

2 The Auxiliary Field, $\mathbf{H}$, and Linear Media

1. A coaxial cable consists of two very long cylindrical tubes, separated by linear insulating material of magnetic susceptibility χ_m . A current I flows down the inner conductor and returns along the outer one; in each case, the current distributes itself uniformly over the surface (Fig. 1).

- (a) Find the magnetic field in the region between the tubes. (b) As a check, calculate the $\mathbf{M}$ and the bound currents, and confirm that (together with the free currents) they generate the correct field.

- What is the *net* bound current down the wire?